Bohrium User Guide for AI Competitions

Updated on June 7, 2026

Major Update:

· **Section 1.2** – Added clarification to avoid issues related to the “https://” protocol.

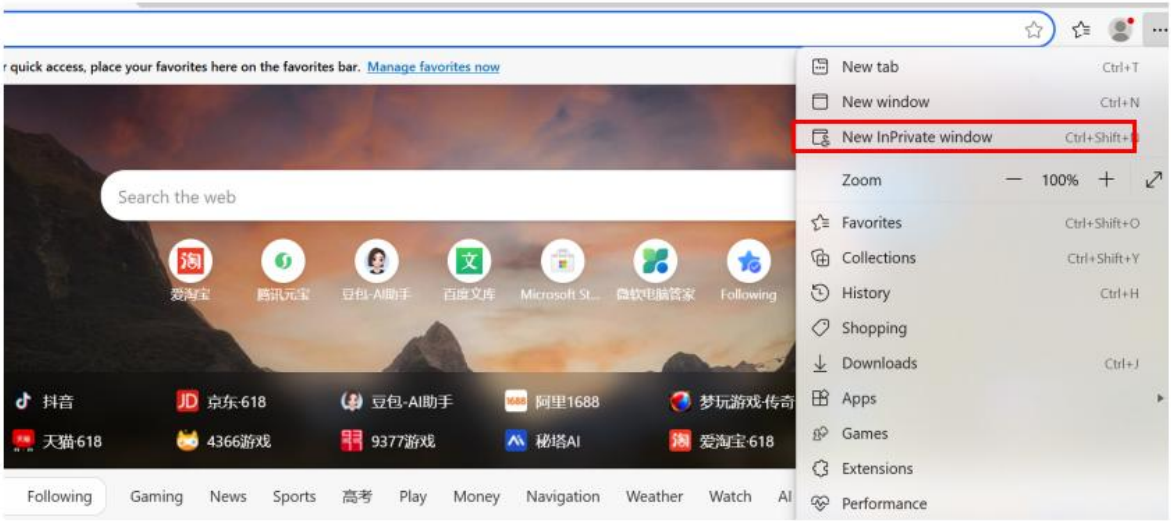
· **Section 2.5** – Added notes on the scope and limitations of using local development environments (e.g., VS Code).

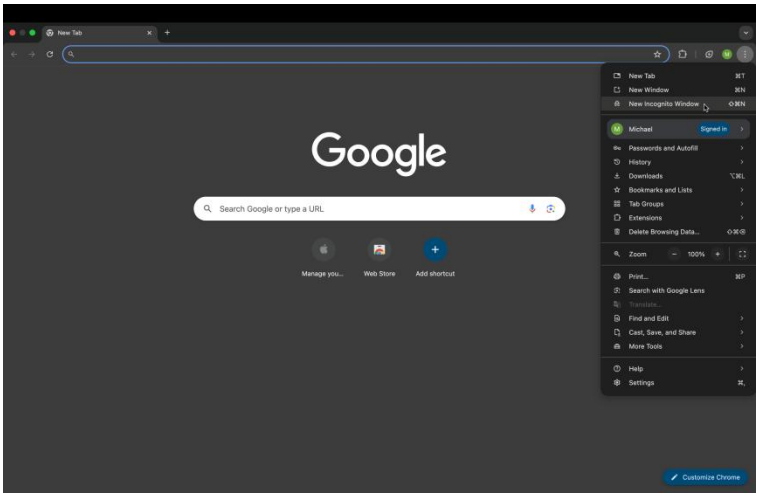
· **Section 3.4** – Added a note advising users **not to** access notebooks through the **Files tab**.

· **Section 8.6** – Added clarification that the Bohrium discussion forum is only used to publish message by the organizers.

1 Login

1.1 Open the Chrome or Edge browser, click “New InPrivate Window” (or “New Incognito Window”) at the top right. Failure to use an InPrivate/Incognito window may result in login issues.

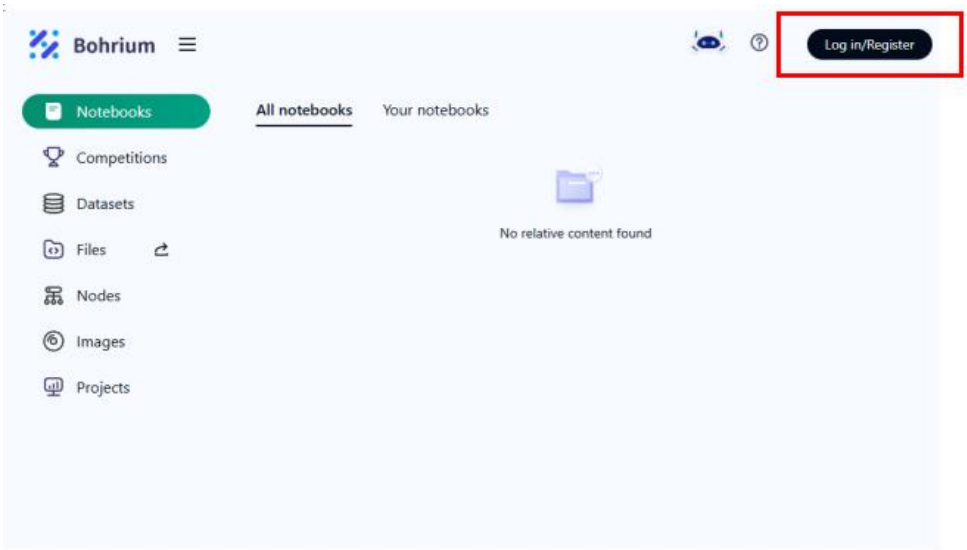




1.2 Enter the URL: https://ioai.bohrium.com/

A small number of browsers do not add “https://” automatically, please guarantee add “https://”.

Click the button in the top right corner to “Log in/Register”.



1.3 Enter your provided login credentials



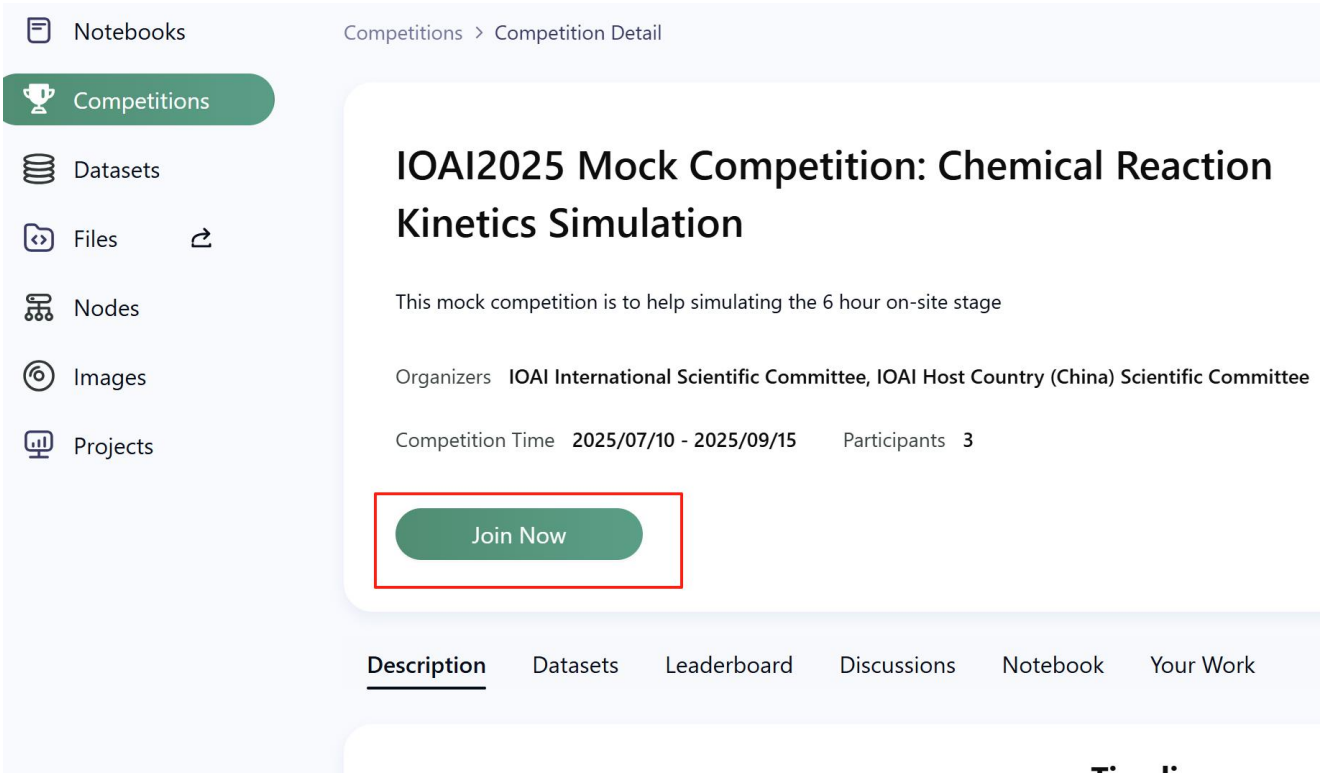
Enter the account and password provided by the contest organizers

The language can be selected as English or Chinese **before login**. When you have already logged in, you must log out first to change the language.

1.4 Join the competition

Please click "Join Now" to enroll in **all competition** tasks before proceeding. Then, you can mount the datasets to the GPU or CPU. Otherwise, the notebook may encounter an error because it cannot access the datasets. You may encounter an error similar to: “FileNotFoundError: TRAIN_PATH = "xxx" # The address of the training

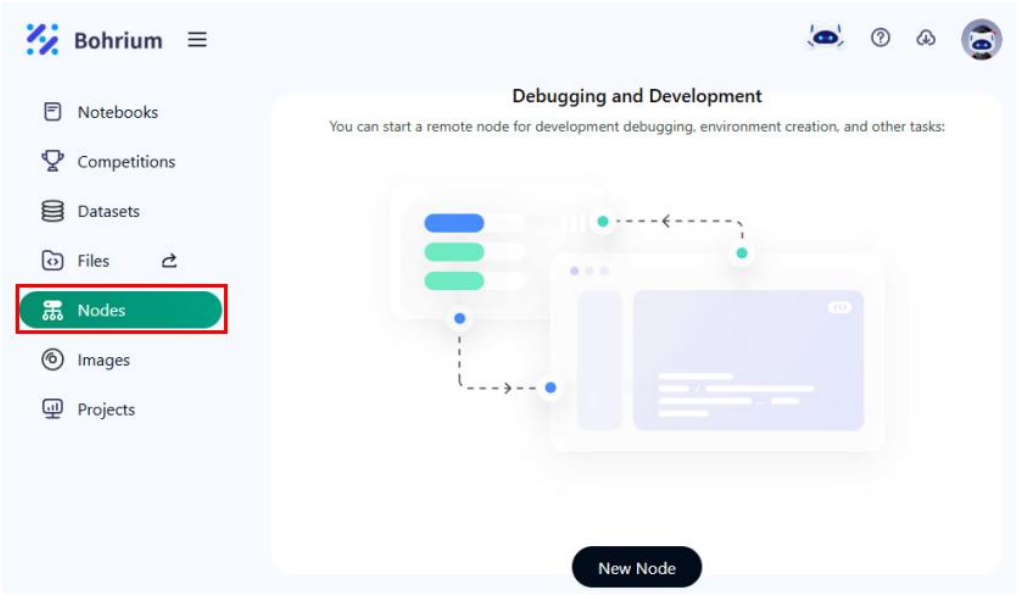
set....”



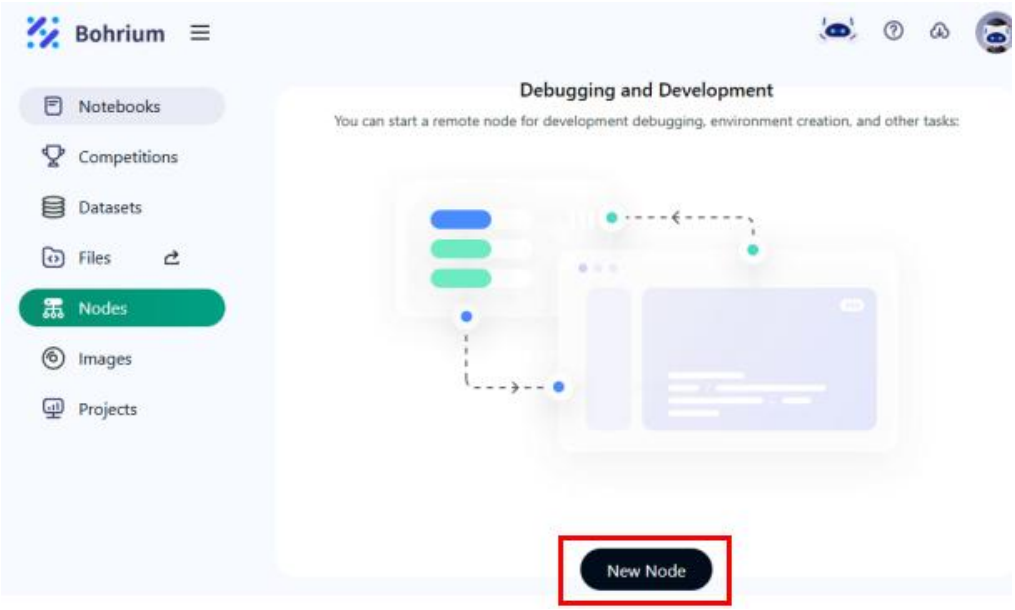
2 Create and Start a Node

This step is for connecting to your personal GPU or CPU node.

2.1 Click on “Nodes” on the left panel



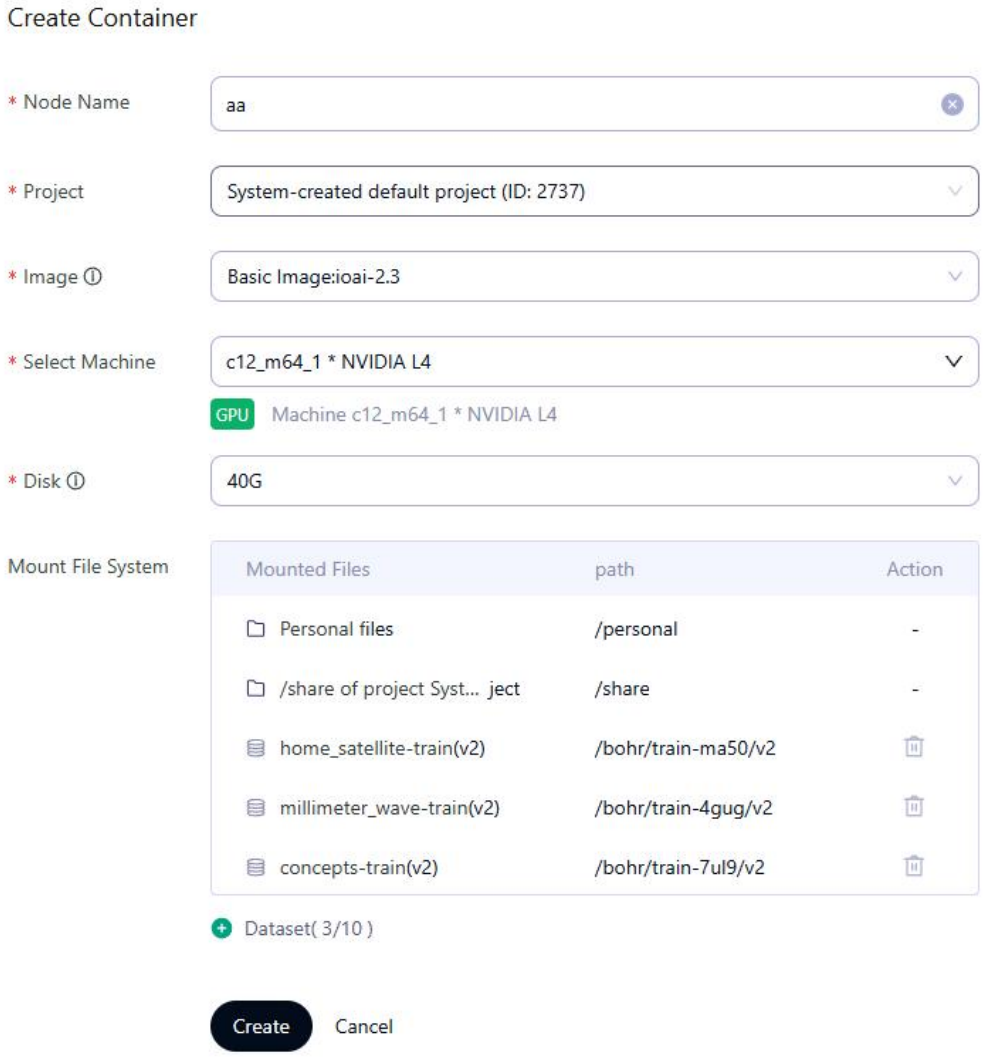
2.2 Click the “New Node” button



2.3 Enter node information

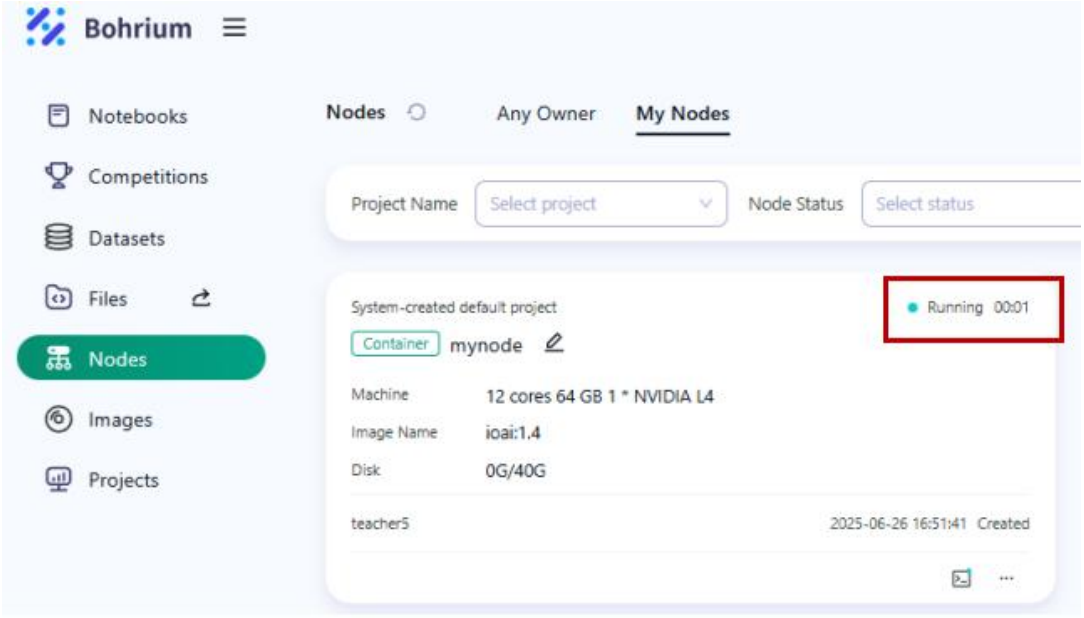
Node Information:

- Node Name: Choose a custom name, 2–128 characters long. It cannot start with special characters or numbers. It can only contain the following special characters: period (.), underscore (_), hyphen (-), and half-width colon ().
- Project: Select the System-Created Default Project (ID: xxxx)
- Image: Select the Public Image - ioai-xxxx from the upper section
- Select Machine: Select c20_m80_1 (NVIDIA L20, 80 GB) for GPU workloads, or c4_m8_cpu for CPU workloads.
- Disk: Select 40G
- Datasets: Select the datasets with the latest versions for each task. If you need to mount your own datasets, refer to section 5.



2.4 After selecting the options, click “Create Node.” Then wait for the node to start up completely (it will display “Running” once it’s ready). Your machine will be guaranteed

to start within 5 minutes. Each account may occupy only one GPU or CPU node at a time. When the contestants want to change or refresh the node, please delete the current one.



2.5* SSH is not supported.

Since both the GPU and CPU computing environments are hosted within Bohrium internal infrastructure, SSH access is not available.

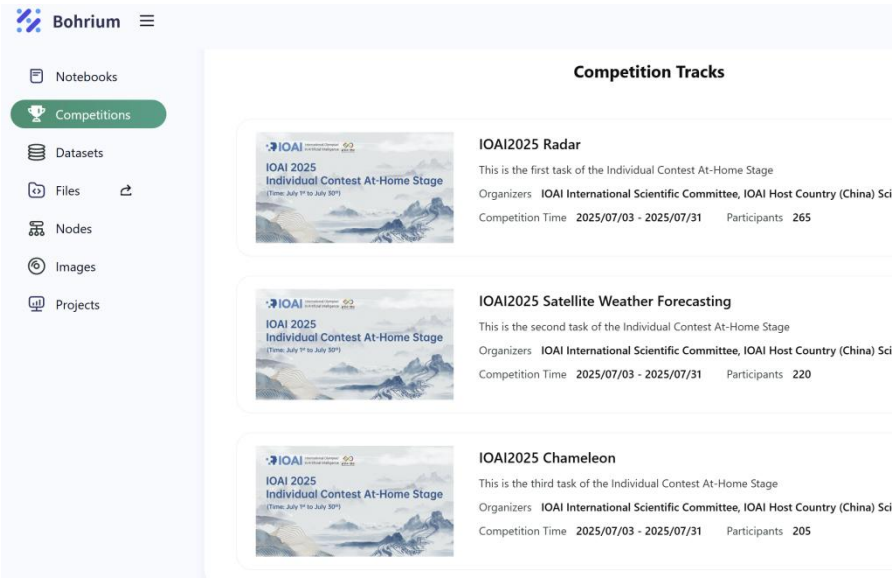
For a more efficient development experience, we recommend using a local development environment (e.g., VS Code) for coding, debugging, and version control.

Once your implementation is complete, you can upload or copy the finalized code into a Bohrium Notebook for execution and submission.

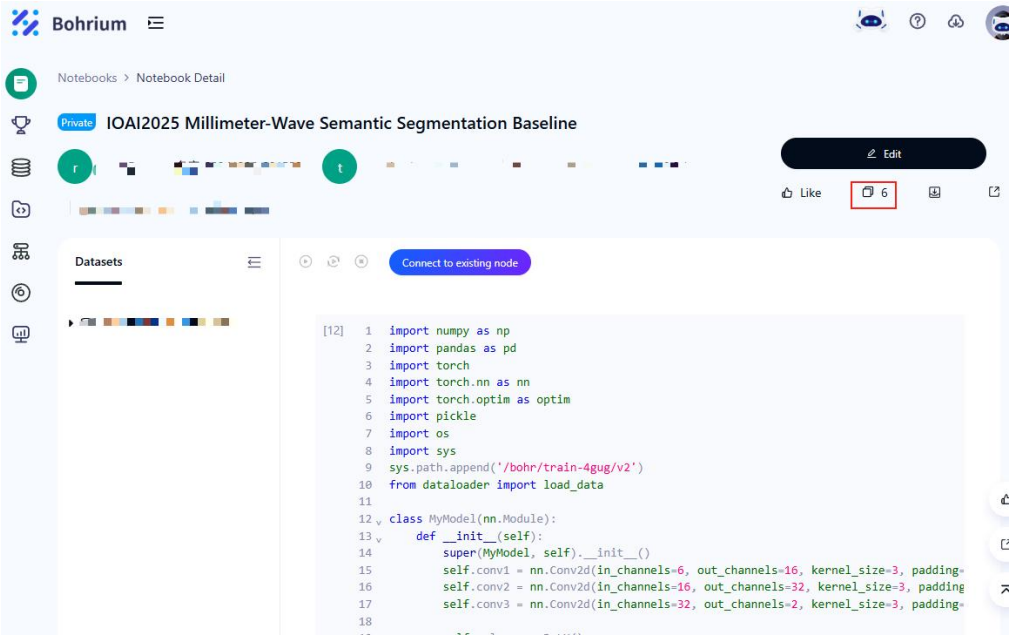
It is not allowed to use other LLMs or auto-coding tools in local development environment (e.g., VS Code) or connect other websites except Bohrium.

3 Run Your Code in a Bohrium Notebook

3.1 Enter the competition interface and read the task description

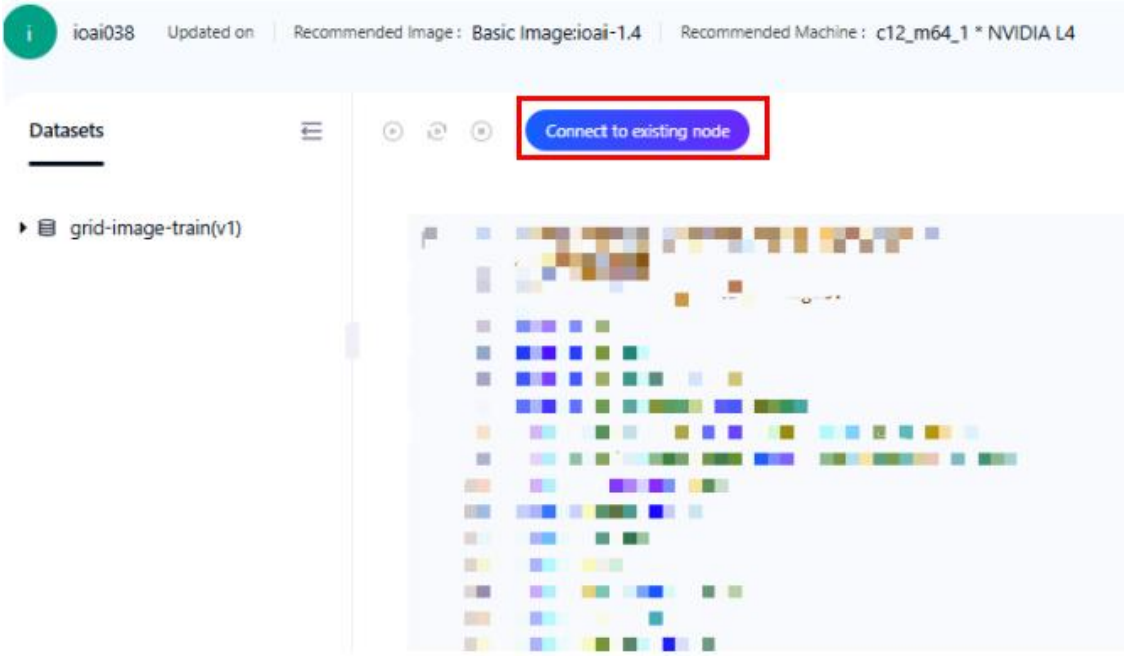


After reading the competition instructions, select the problem and open “baseline.ipynb” (a link should be provided in the problem statement). Once inside, click the button shown below to create a “copy”, and you can start training your own model.



Note: It is recommended to create a new notebook copy whenever significant changes are made. For the copy name, it is suggested to use "APOAI_QuestionNumber_vVersionNumber" to easily distinguish submission versions. The copy name must not contain any line breaks, otherwise it will raise an error.

3.2 After creating the copy, click "Connect to existing node" to connect to the GPU/CPU and then run the notebook.



3.3 Note: Please remember to free memory when multiple notebooks are connected to the same GPU/CPU node. Otherwise, you may encounter out-of-memory (OOM) errors.

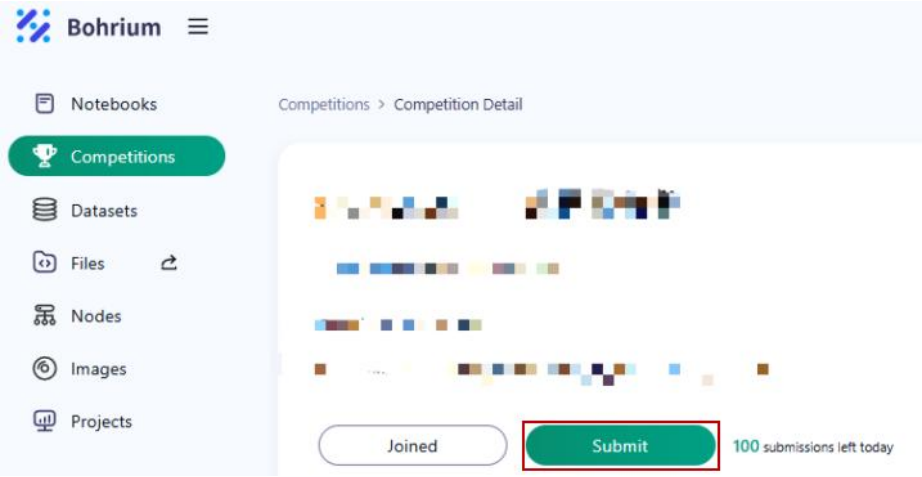
The command below is usually used to clean memory.

```
gc.collect()
torch.cuda.empty_cache()
```

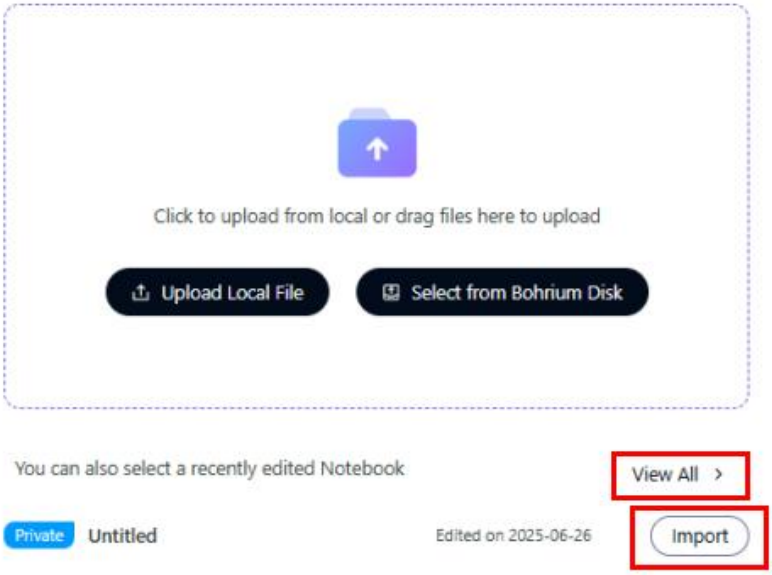
3.4 Note: Please make sure to open notebooks using the "Notebook" button on the left sidebar, rather than through the "Files" panel.

4 Submit Your Notebook

After the notebook debugging is complete, return to the competition interface and click “Submit” to submit your solution.



The notebook you just used will appear at the bottom of the interface. Click “Import” to complete the submission.



If you cannot find the notebook, click “View All” and select your own notebook file. The testing machine will run the entire notebook from the beginning to the end, including the training process.

Based on data from previous competitions, some solutions may require 10–20 minutes to run on the evaluation server when large datasets are used.

4.1 Notebook Size Limit

The maximum size of the Submission notebook is 100 MB

4.2 If your submission fails with the error message “Post failed, please retry,” please create a new notebook copy and complete the submission process again.

This issue is typically caused by one of the following reasons:

- The notebook execution was interrupted before completion, causing some data or metadata to be saved incompletely.
- The notebook filename contains invalid characters, such as line breaks, carriage returns, or other unsupported special characters.

5 Using Custom Datasets and Pretrained Models

If you want to include a trained model (.pth) file or upload your own datasets, you need to create new “Datasets.”

“Datasets” are basically directories in the Bohrium cloud.

You may store your model weights files in your created datasets and access them in your submitted notebook.

5.1 Save the trained model

You can use the following command in the notebook to save the trained model parameters to the “Files” on the left panel of the platform:

```
torch.save(model.state_dict(), ‘xxxmodel.pth’)

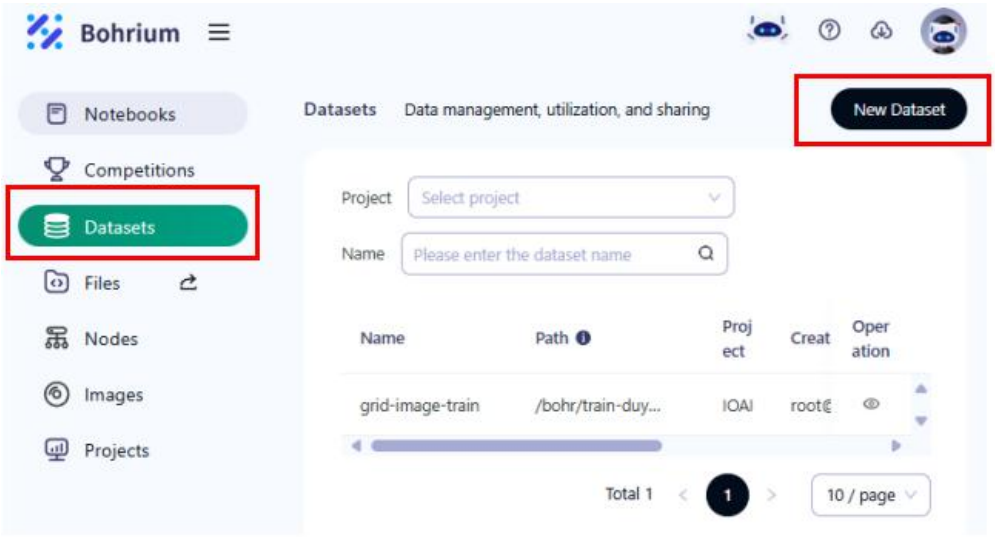
!cp xxxmodel.pth /personal    # Move the model file to /personal.
```

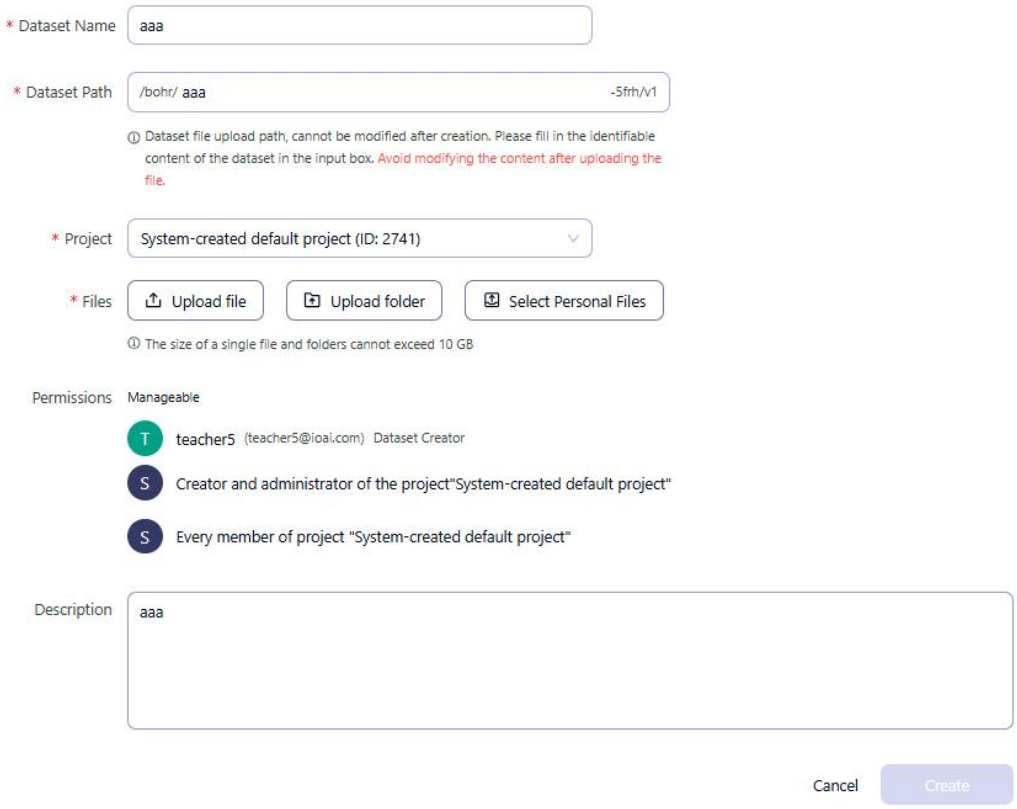
The /personal directory is for storing your files and will be synced to all of your runtimes on Bohrium.

5.2 Create new “Datasets”

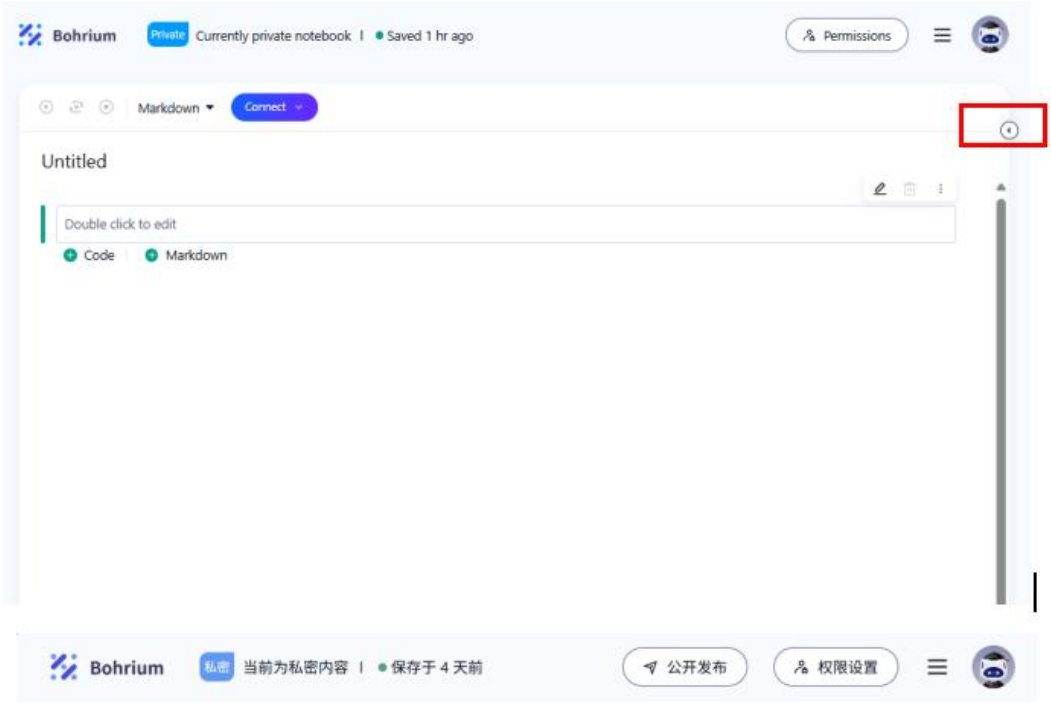
Create “Datasets” as shown in the figure below, and upload your own model by uploading “Files” into the datasets.

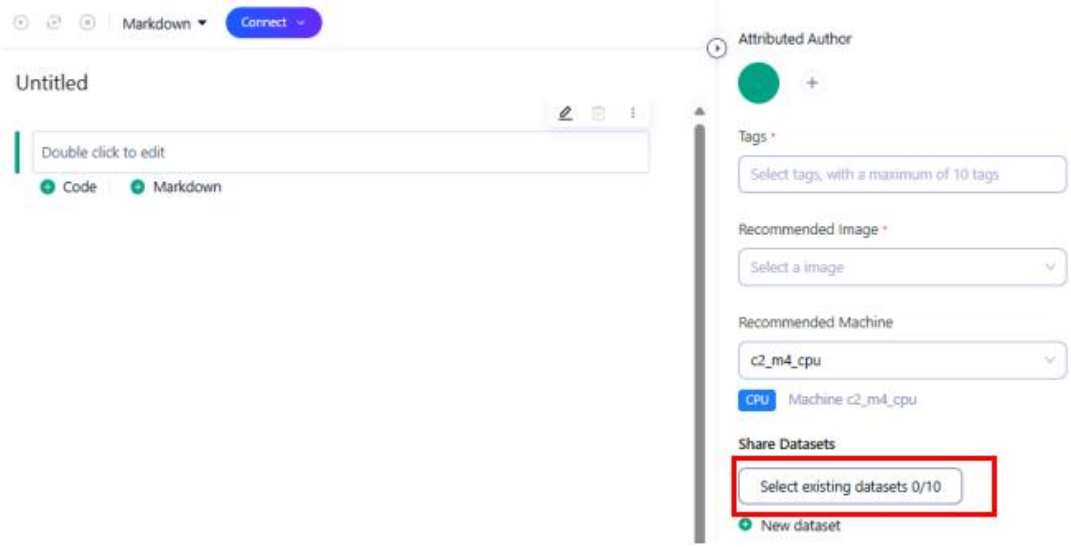
You can also upload datasets from your laptop. The datasets are usually created instantly, and you can refresh the page to see the changes.


Bohrium
Notebooks
Competitions
Datasets
Files
Nodes
Images
Projects
Datasets
Data management, utilization, and sharing
New Dataset
Project
Select project
Name
Please enter the dataset name
Name
Path
Project
Creat
Operation
grid-image-train
/bohr/train-duy...
IOAI
root@
Total 1
1
10 / page


* Dataset Name
aaa
* Dataset Path
/bohr/ aaa
-5frh/v1
Dataset file upload path, cannot be modified after creation. Please fill in the identifiable content of the dataset in the input box. Avoid modifying the content after uploading the file.
* Project
System-created default project (ID: 2741)
* Files
Upload file
Upload folder
Select Personal Files
The size of a single file and folders cannot exceed 10 GB
Permissions
Manageable
T
teacher5 (teacher5@ioai.com) Dataset Creator
S
Creator and administrator of the project"System-created default project"
S
Every member of project "System-created default project"
Description
aaa
Cancel
Create

5.3 Reading your model weights file:





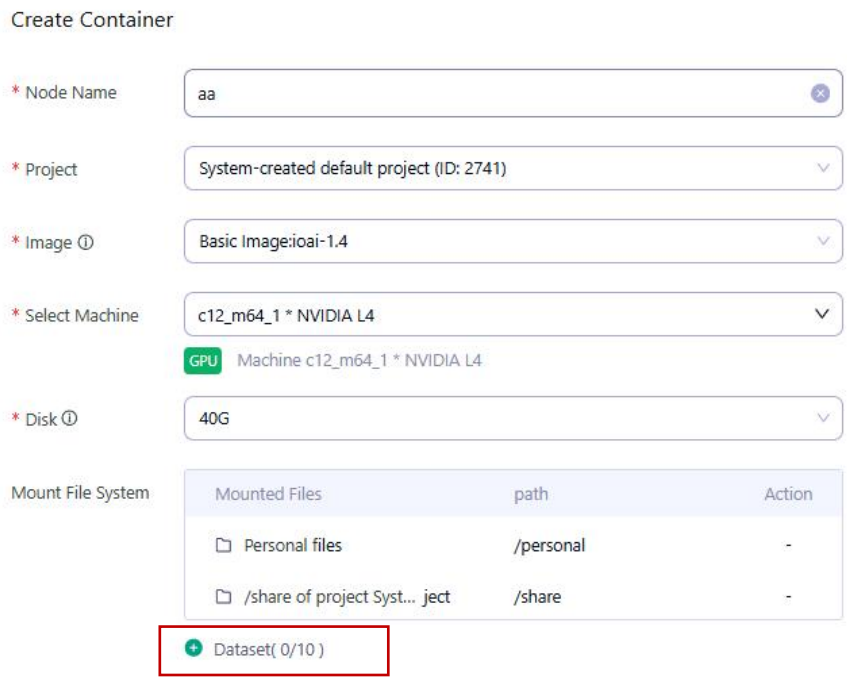
Open the notebook you intend to submit, and on the right panel, click on “Select existing datasets,” then attach the datasets that you want to use. At this point, you can click “Copy datasets address” to copy the “Datasets” URL and paste it into the notebook. This is the direct path from which you can read the contents of the datasets. Don’t forget to save the notebook.

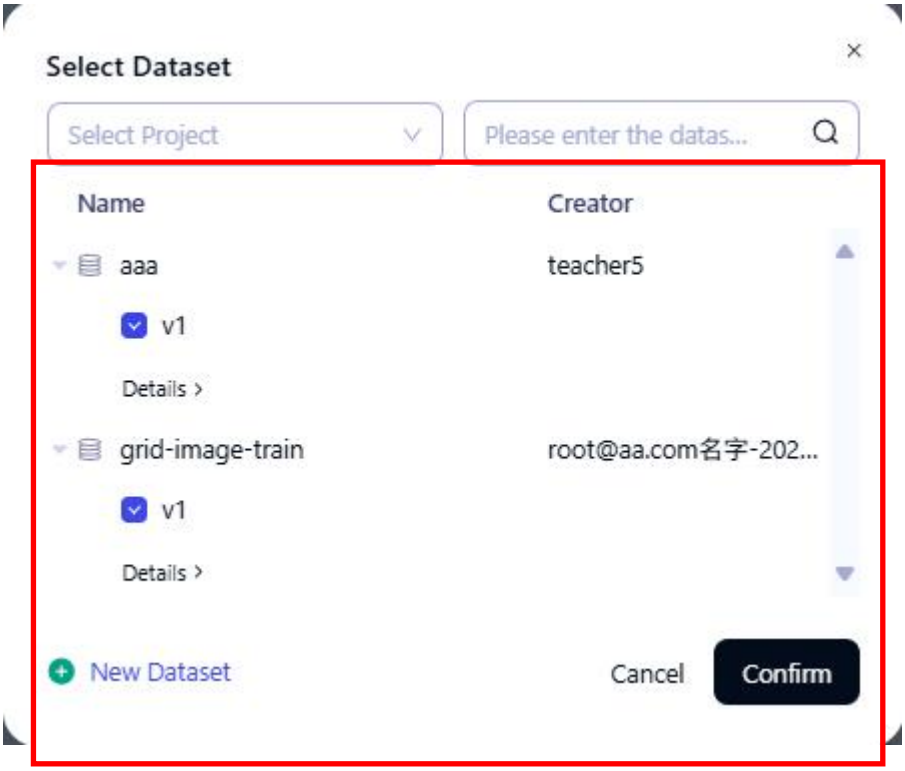
5.4 At this point, if you directly run the notebook according to section 5.3, you will encounter an error. This is because your GPU/CPU node cannot connect to the datasets.

Your node will only have access to the files and datasets that you selected to mount during the node creation process. You will need to recreate a node according to Section 2 and remember to mount your own datasets on the machine.

To avoid unnecessary trouble, you can also mount all “Datasets” that you can see when creating a node, but this may increase the node startup time.

Here is a step-by-step guide for mounting datasets onto your nodes.





Click on the arrow icon on the left of each dataset to view the versions and select the version that you want to mount.

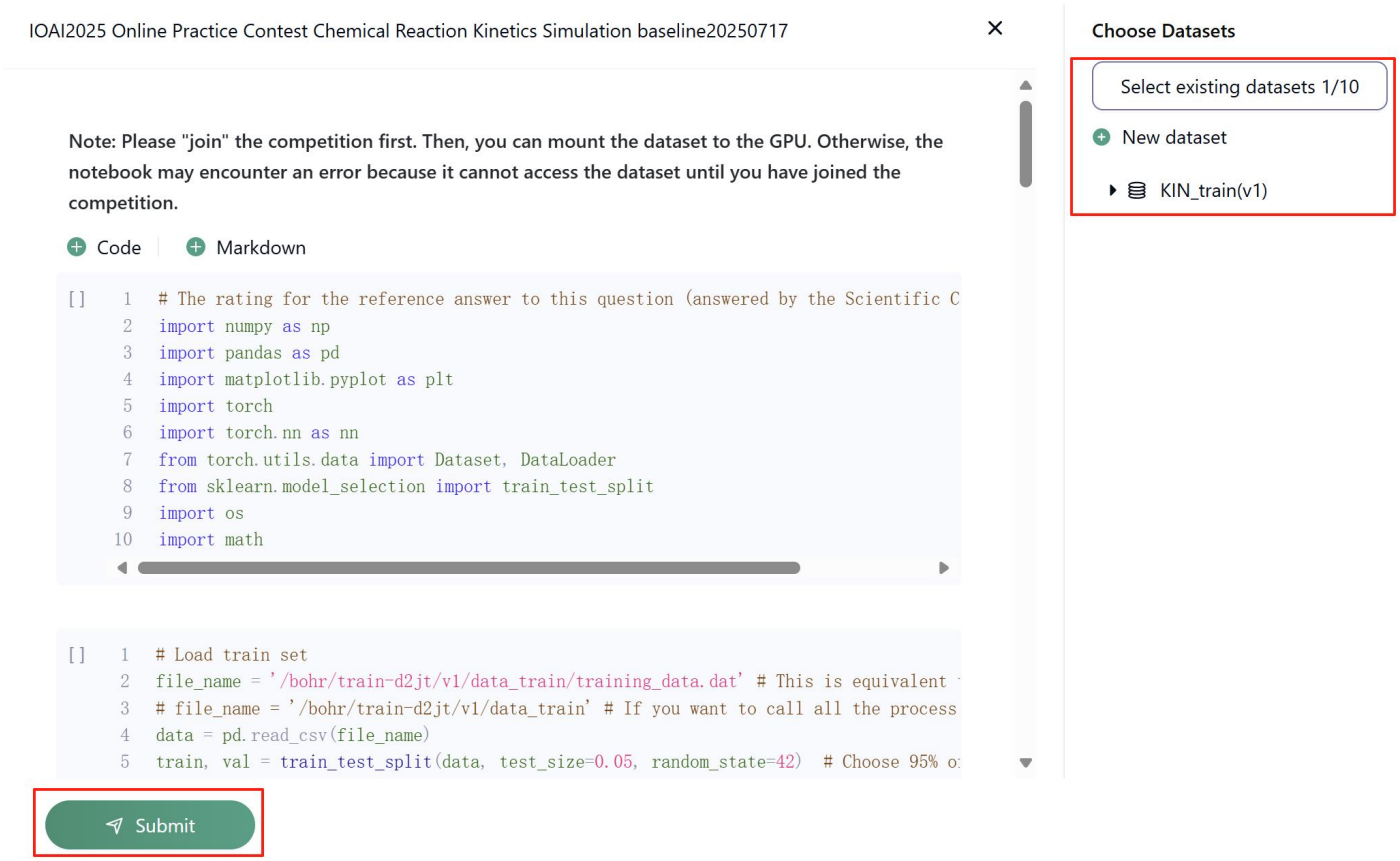
After restarting the machine, go back to Section 5.3. You will find that your notebook can now import your own “datasets” and run successfully.

When you submit the notebook, if your datasets have been attached to the submission notebook according to 5.3, they will be automatically submitted to the testing machine along with the notebook.

5.5 Manually Attaching Datasets to a Submission

If the datasets are not automatically attached, you may add or remove datasets manually before submission.

Use the datasets management panel in the upper-right corner to attach your own datasets, model weights (.pth files), or other required files.



Note: Some tasks may allow you to submit only one notebook. In this case, you cannot mount your own datasets and must remove all datasets. Otherwise, you will receive an error message at the top:

“In this task, participants can only submit a notebook and cannot submit any mounted datasets or .pth files generated by themselves. Please check the upper-right corner to remove the mounted datasets first.”

6 Leaderboard

6.1 Leaderboard A and Leaderboard B

The final ranking is determined by Leaderboard B.

Leaderboard A is provided as a public reference during the competition and is intended to help contestants evaluate their current performance. It displays the best score achieved by each contestant on the public evaluation set.

6.2 Selecting Submissions for Leaderboard B

Each contestant may manually select up to two successful submissions for inclusion in Leaderboard B. The final score for a task will be the higher score among the submissions selected for Leaderboard B.

6.3 Automatic Selection

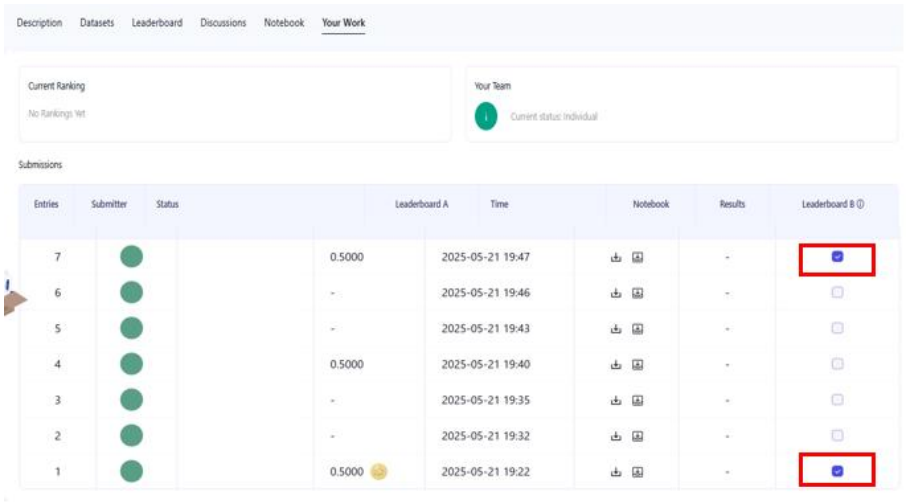
If no successful submission is manually selected for Leaderboard B, the system will automatically choose the submission with the best score on Leaderboard A. The selected submission will then be used to calculate the contestant’s Leaderboard B score.

6.4 Tie-breaking Rule

If multiple eligible submissions have the same Leaderboard A score, the most recent submission will be selected automatically.

6.5 Submission Requirements

If you have at least one successful submission, at least one submission must be selected for Leaderboard B. An empty selection is not allowed.



7 AI assistant

During **APOAI 2026,** contestants will have access to ChatGPT-4o-mini through Bohrium. Each request is limited to 1,024 tokens. Click the icon shown below to open the assistant. Depending on your browser and screen layout, the icon may appear in either the upper-right or lower-right corner of the interface.

The assistant window can be freely repositioned by dragging the icon.

 or 

Unless explicitly permitted in the documentation, only the large language models built into Bohrium may be used. The use of any external platforms, libraries, or models is prohibited. This includes, but is not limited to, scikit-learn and PyTorch.

8 FAQ

8.1 Can I use another editor instead of the Bohrium notebook editor?

A. Yes. We recommend developing and debugging your code locally using VSCode or another editor, then copying the final code into a Bohrium notebook for execution and submission.

8.2 How can I check the Python and pip environment in Bohrium?

A. Please connect a notebook to the node and then use “!pip --version” to check the version and use “!pip list” to check the installed packages.



```
1  !pip list
```

Run successful

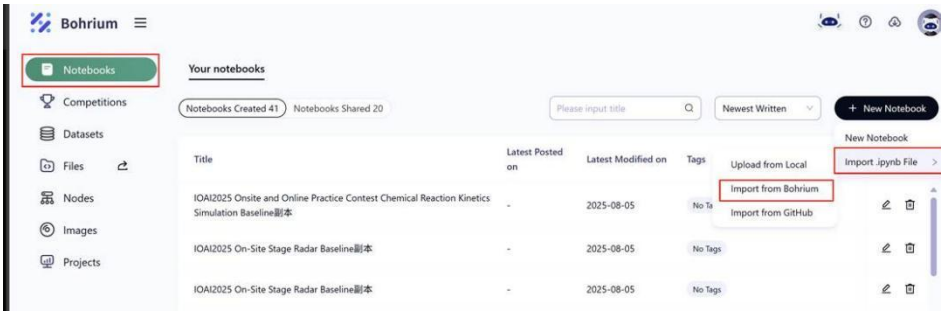
Package	Version
absl-py	2.3.1
accelerate	1.8.1
aext-assistant	4.1.0
aext-assistant-server	4.1.0
aext-core	4.1.0
aext-core-server	4.1.0
aext-panels	4.1.0
aext-panels-server	4.1.0
aext-project-filebrowser-server	4.1.0
aext-share-notebook	4.1.0
aext-share-notebook-server	4.1.0
aext-shared	4.1.0
aiobotocore	2.19.0

8.3 Can I download and use Python libraries in the notebook that are not installed?

A. No.

8.4 How do I open a notebook stored in the /personal directory?

Open the “notebook” tab. In the top right corner, “notebook”->“Import .ipynb File”->“Import from Bohrium”.



8.5 Which pretrained models are available in torchvision.models?

Only the 3 pretrained ResNet parameters are available.

- ResNet18, ResNet34, ResNet50

Usage example for ResNet18:

from torchvision.models import resnet18, ResNet18_Weights

model = resnet18(weights=ResNet18_Weights.DEFAULT)

```

8.6 Can students ask questions through the Bohrium discussions forum?

A: No. The discussion forum is used exclusively for official announcements. In most cases, announcements will also be posted in the WhatsApp group and relayed to contestants by their respective team leaders.
```